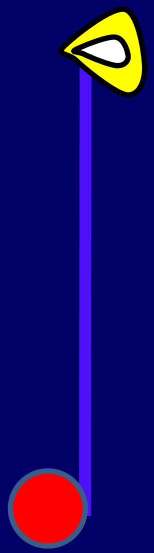
Quiz 6: The problem is related with Yo-Yo which is the most favorite game for pupils. Assuming Yo-Yo is a weightless line winding a disk of the radius *R* and the mass *m*.

a. If we hold the end of the line to be steady, calculate the acceleration of the disk and the force to act on the end of the line.

b. If we want the disk to maintain a fixed height, how large lift force is needed and what is the angular acceleration?

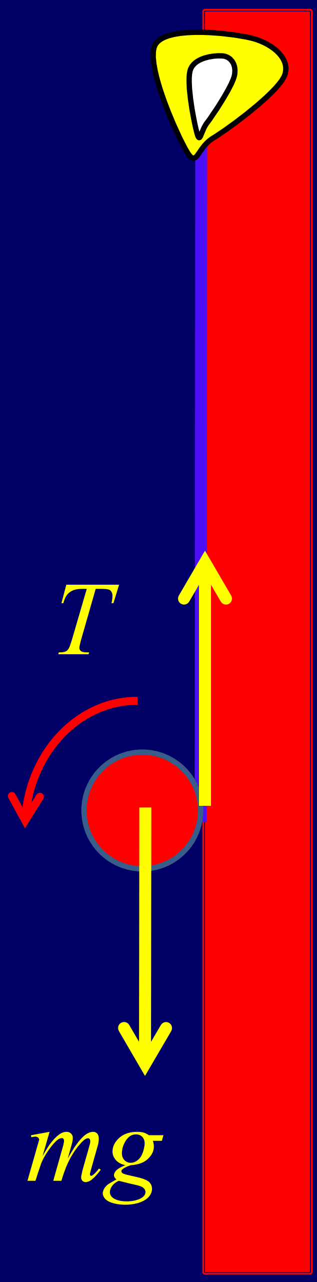
c. If the lift force is *2mg*, what is the acceleration of the line end that we hold?



a. Rolling without slipping $a_c = R\alpha$

b. Pure rolling $a_c = 0$

c. Rolling with slipping $a_c \neq R\alpha$



Solution: a.

$$mg - T = ma_c \quad ---(1)$$

$$TR = I\alpha = \frac{1}{2}mR^2\alpha \quad ---(2)$$

$$\alpha = \frac{2g}{3R} \qquad a_c = R\alpha = \frac{2g}{3}$$

From (1), we have $T = \dfrac{1}{3}mg$

b.

$$mg - T = 0 \quad ---(3)$$

$$TR = I\alpha = \frac{1}{2}mR^2\alpha \quad ---(4)$$

$$\alpha = \frac{2g}{R}$$

c.

$$T - mg = ma_c \quad ---(5)$$

$$TR = \frac{1}{2}mR^2\alpha \quad ---(6)$$

$$\alpha = \frac{4g}{R}$$

$$a = \alpha R + a_c = 5g$$